Basic concepts of Probability (2)

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Probability Laws

- Two basic rules/laws of probability theory-

1. Additive rule
2. Multiplicative rule

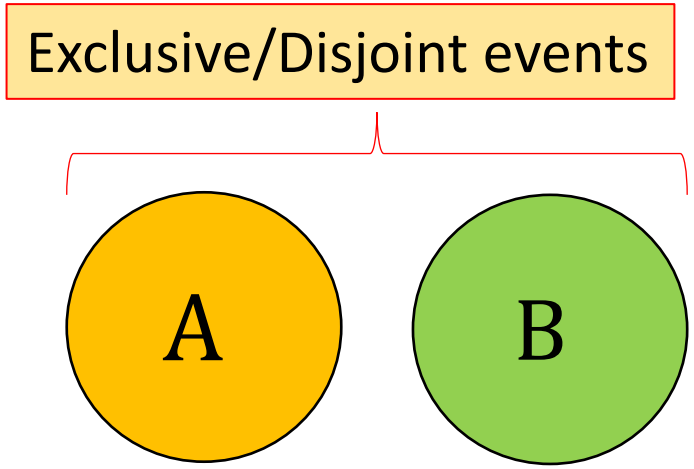
Disjoint, Joint events

Independent, Dependent events

Additive rule

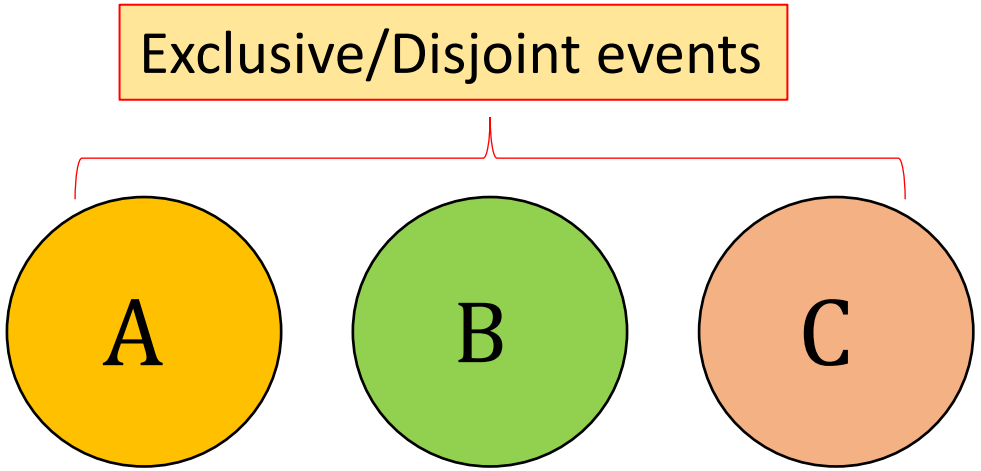
For disjoint events:

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$



For disjoint events:

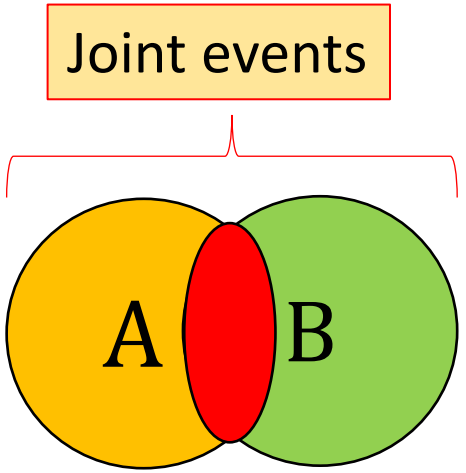
$$P(A \text{ or } B \text{ or } C) = P(A \cup B \cup C \ldots)$$
$$= P(A) + P(B) + P(C)$$



Additive rule

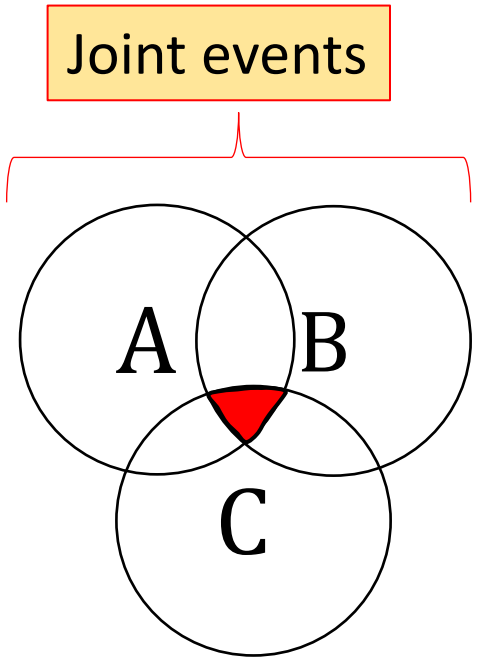
For joint events:

$$P(A \text{ or } B) = P(A \cup B)$$
$$= P(A) + P(B) - P(A \cap B)$$



For joint events:

$$P(A \text{ or } B \text{ or } C) = P(A \cup B \cup C)$$
$$= P(A) + P(B) + P(C) - P(A \cap B)$$
$$- P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$



Additive rule

Example: In a class, 70% of the student good at math, 30% good at chemistry and 10% at both. What is the probability that the student good at either pen or pencil?

Solution: $P(M) = 0.7, P(C) = 0.3, P(M \cap C) = 0.1$

$\therefore P(M \text{ or } C) = P(M \cup C) = P(M) + P(C) - P(M \cap C)$

$$= 0.7 + 0.3 - 0.1 = 0.9$$

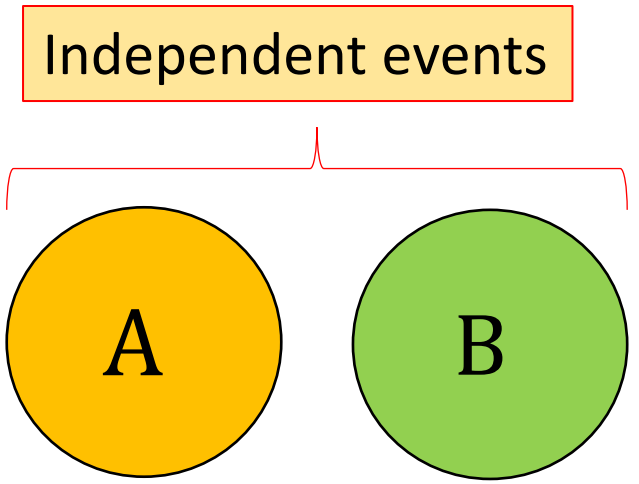
Additive rule

- Example: Mr. Ali feels that the probability that he will pass mathematics is 2/3 and statistics is 5/6.If the probability that he will pass both the course is 3/5, what is the probability that he will pass at least one of the course? (Ans: $\frac{9}{10}$)

Multiplicative rule

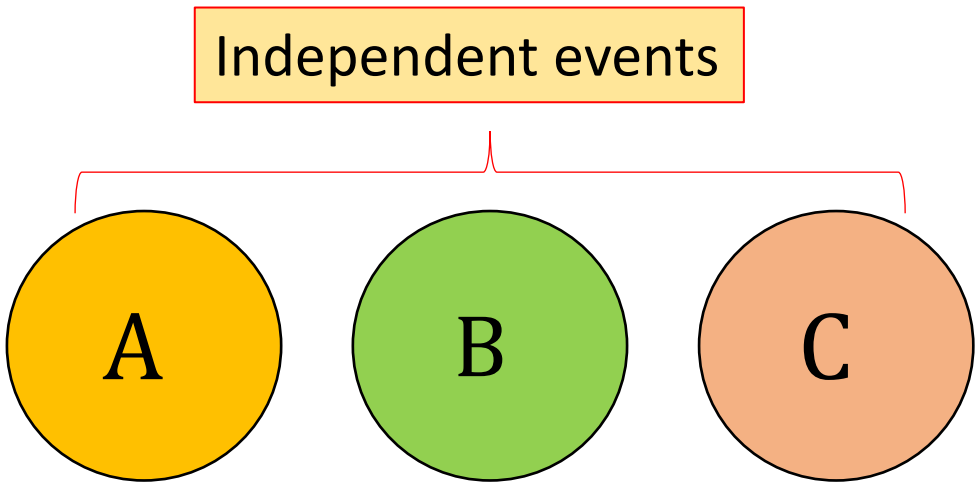
For independent events:

$$P(A \,\&\, B) = P(A \cap B) = P(A) \times P(B)$$



For independent events:

$$P(A \,\&\, B \,\&\, C) = P(A \cap B \cap C \ldots)$$
$$= P(A) \times P(B) \times P(C)$$



Multiplicative rule

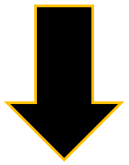
- If the probability that student A will be fail is 0.7 and the probability that student B will fail is 0.6. What is the probability that they will both fail?

- These are independent events
- $P(A\ and\ B) = P(A \cap B) = P(A) \times P(B) = 0.42$

Multiplicative rule

For dependent events:

$$P(A \,\&\, B) = P(A \cap B) = P(A|B)\,P(B)$$



Conditional probability



Probability of A “given that” B

Conditional Probability

- Conditional probability refers to the chances that some outcome occurs given that another event has also/already occurred.

- Let, $A = One\ event, B = Another\ event\ already\ occured$

- It is often stated as the probability of A given B and is written as $P(A|B)$, where the probability of A depends on that of B happening.

$$P(A|B) = \frac{P(A\ and\ B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability

- Example: In a class, 70% of the student good at math, 30% good at chemistry and 10% at both.

- If it is known that the student good at math, then what is the probability that the employee also good at Chemistry?

- $P(C|M) = \frac{P(M \cap C)}{P(M)} = \frac{0.1}{0.7} =$

Conditional Probability

- Suppose a balanced die is rolled once.
  a) Find the probability that a number divisible by 3 is rolled given that the die comes up even.
  b) Find the probability that the die comes up even given that a number divisible by 3 is rolled.
  c) Find the probability that a number divisible by 3 is rolled given that die comes up at most 4.
  d) Find the probability that the die comes up at most 4 given that a number divisible by 3 is rolled.

Conditional Probability

Sample space for rolling a die once, $S = \{1, 2, 3, 4, 5, 6\}$

a) Let,

$A = Number\ divisible\ by\ 3 = \{3, 6\}$

$B = Even\ number = \{2, 4, 6\}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\therefore P(A|B) = \frac{\frac{1}{6}}{\frac{1}{2}} = 0.33$$

Probability that a number divisible by 3 is rolled given that the die comes up even.

$$A \cap B = \{3, 6\} \cap \{2, 4, 6\} = \{6\}$$

$$P(A \cap B) = \frac{1}{6}$$

$$B = \{2, 4, 6\}$$

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

Conditional Probability

b) $(Ans: 0.5)$

c) $(Ans: 0.25)$

d) $(Ans: 0.5)$

Conditional Probability

1. In rainy season, it rains 60% of the days in Bangladesh. When it rains, 80% times it makes thunderstorms. What is the probability that, in a particular day of rainy season, it will rain and it will thunderstorm?

R be an event of rain

T be an event of Thunderstorms

$$P(R \text{ and } T) = P(R \cap T)$$

$P(R) = 60\% = 0.6$

$P(T|R) = 80\% = 0.8$

$$P(T|R) = \frac{P(T \cap R)}{P(R)}$$

$$P(T \cap R) = P(T|R) \times P(R)$$

$$P(T \cap R) = 0.8 \times 0.6 = 0.48$$

Self Practice

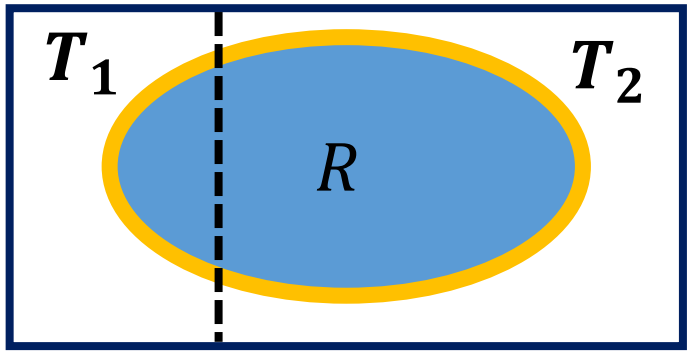
1. In a class, 70% of the student are good at math, 30% good at chemistry and 10% at both. Then randomly select 1 student.

a) What is the probability that if student is good at math, he or she will be good at chemistry?

b) What is the probability that if student will not be good at math?

c) What is the probability that if student is not good at math, he or she will be good at chemistry

Bayes' Theorem

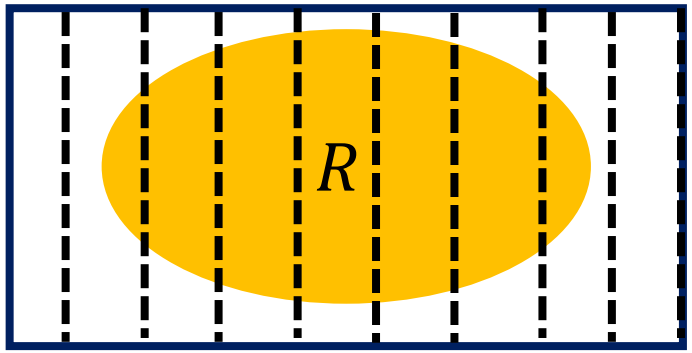


- Let, events T_1 and T_2 form partition of S. Let R be an event with $P(R) > 0$. Then,

$$P(T_1|R) = \frac{P(T_1 \cap R)}{P(R)} = \frac{P(T_1)P(R|T_1)}{P(T_1)P(R|T_1) + P(T_2)P(R|T_2)}$$

$$P(T_2|R) = \frac{P(T_2 \cap R)}{P(R)} = \frac{P(T_2)P(R|T_2)}{P(T_1)P(R|T_1) + P(T_2)P(R|T_2)}$$

Bayes' Theorem



- Let, events $T_1, T_2, \ldots$ form partition of S. Let R be an event with $P(R) > 0$. Then,

$$P(T_j|R) = \frac{P(T_j \cap R)}{P(R)} = \frac{P(T_j)P(R|T_j)}{P(T_1)P(R|T_1) + P(T_2)P(R|T_2) + \cdots}$$

Example

- 60% of the students in a class are male. 20% of the males and 10% of the females are absent in a particular day if a student ID is randomly selected from the attendance.

a. What is the probability that the student is absent?

b. If the randomly selected student is absent, what is the chance that the student is female?

Solution

$60\%\ of\ the\ students\ in\ a\ class\ are\ male$

$40\%\ of\ the\ students\ in\ a\ class\ are\ female$

$20\%\ of\ the\ males\ are\ in\ the\ photography\ club$

$10\%\ of\ the\ females\ are\ in\ the\ photography\ club$

- $P(M) = 0.6$
- $P(F) = 0.4$
- $P(A|M) = 0.2$
- $P(A|F) = 0.1$

a) What is the probability that the student is absent?

$$P(A) = P(M)P(A|M) + P(F)P(A|F) =$$

b) If the randomly selected student is absent, what is the chance that the student is female

$$P(F|A) = \frac{P(F)P(A|F)}{P(A)} =$$

Example

$P(D|B_1) = 0.3$

$P(D|B_2) = 0.9$

- Dr. X diagnoses cancer correctly 80% cases. The chance that a patient will die by his treatment after correct diagnosis is 30%, and the chance of death by wrong diagnosis is 90%. A patient of Dr. X who had cancer died. What is the probability that his diagnosis was wrong?

$P(B_2) = 0.2$

$B_1 = Cancer\ is\ correctly\ diagnosis$

$B_2 = Cancer\ is\ not\ correctly\ diagnosis$

$P(B_1) = 0.8$

$D = Cancer\ patient\ died$

$P(B_2|D) = ????$

$$P(B_2|D) = \frac{P(B_2)P(D|B_2)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2)}$$

Contingency table

- A contingency table, also known as a cross-tabulation or crosstab, is a statistical table used to analyze and display the relationship between two categorical variables.

- For example, the question, "Do you like watching documentaries?" was asked of 100 people. Results are shown in the table.

	Yes	No	Total
Male	22	38	60
Female	12	28	40
Total	34	66	100

Joint Probability

- Both event occur together
- For example, The question, "Do you like watching documentaries?" was asked of 100 people. What is the probability of a randomly selected individual being a male who likes watching documentaries?

	Yes	**No**	**Total**
Male	22	38	60
Female	12	28	40
Total	34	66	100

Marginal Probability

- Single event occurring independently of any other events.
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Example

- Below given a contingency table for Smoking status and Respiratory illness status.

	Yes	**No**	**Total**
Smoker	9860	1470	
Non smoker	6280	12170	
Total			

1. What is the probability that a randomly selected person have Respiratory illness ?
2. What is the probability that a randomly selected person is non smoker?
3. What is the probability that a randomly selected person is both smoker and has Respiratory illness ?
4. If a person is smoker, what is the probability that he also has Respiratory illness ?

Thank You